CONTINUOUS TRAINING SYSTEM

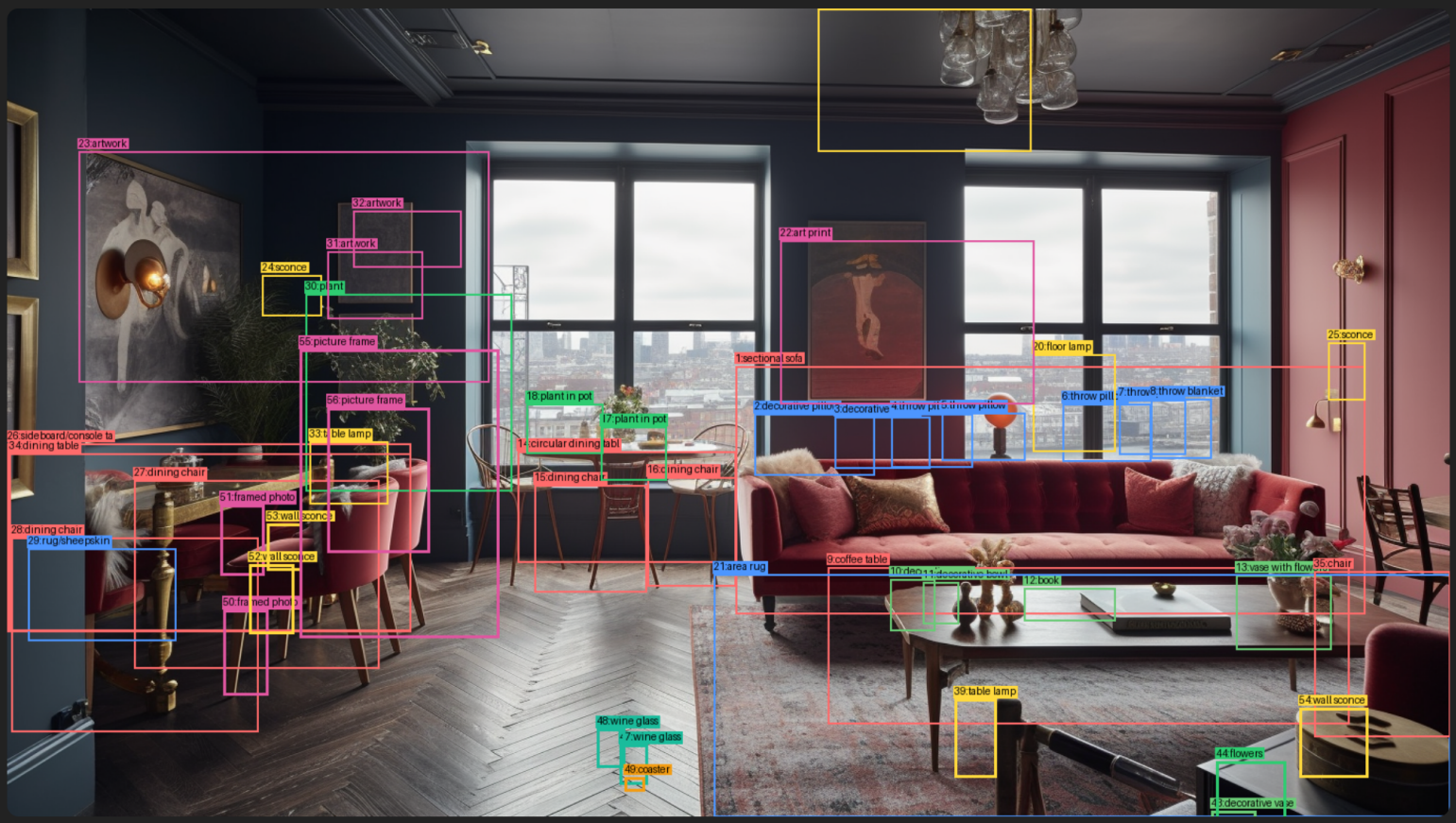
# Home Inventory VLM Architecture

Qwen2.5-VL-3B with QLoRA fine-tuning, incremental learning, and human-in-the-loop correction

| 56 | 9 | 88% | Living Room/Dining Room |
|---|---|---|---|
| TOTAL OBJECTS | CATEGORIES | AVG CONFIDENCE | ROOM TYPE |
| ↑ +21 in pass 2 | | | |

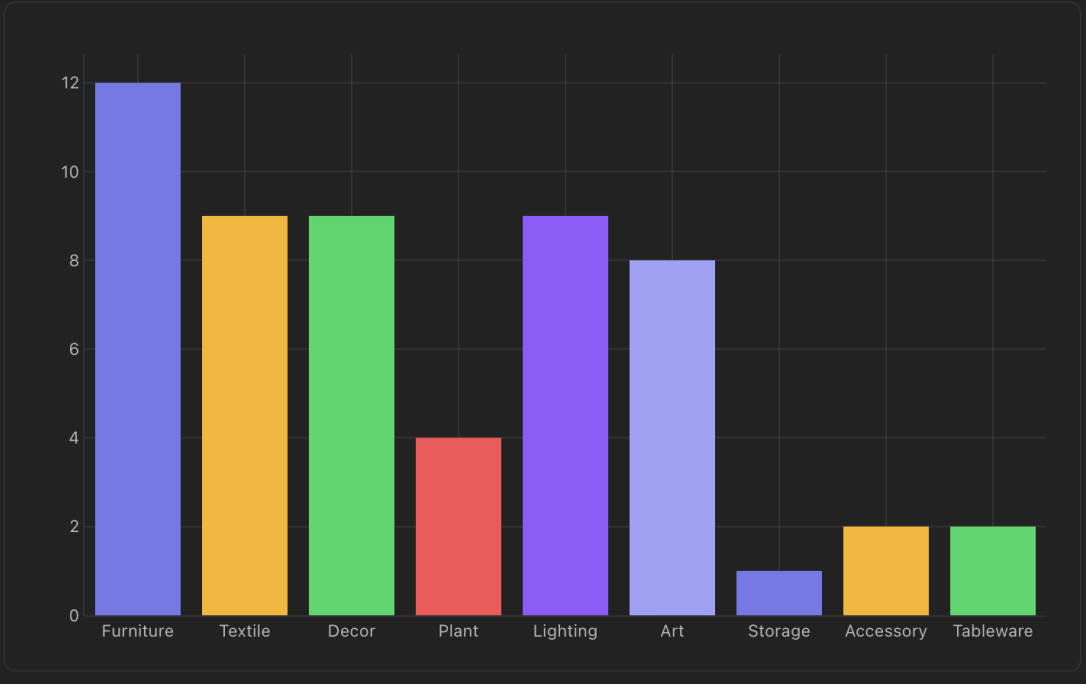
DETECTION

## Annotated Image (Both Passes)




BREAKDOWN

## Items by Category



QUALITY

## Confidence by Pass




INVENTORY

## Complete Item List

Filter...

| ID | PASS | OBJECT | CATEGORY | CONFIDENCE | COLOR | MATERIAL | STYLE | CONDITION | SIZE |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | sectional sofa | Furniture | 98% | red/burgundy | velvet | mid-century modern | good | large |
| 2 | 1 | decorative pillow | Textile | 95% | gold | fabric | contemporary | good | small |
| 3 | 1 | decorative pillow | Textile | 94% | red/burgundy | velvet | contemporary | good | small |
| 4 | 1 | throw pillow | Textile | 93% | red/burgundy | velvet | contemporary | good | small |
| 5 | 1 | throw pillow | Textile | 92% | red/burgundy | velvet | contemporary | good | small |
| 6 | 1 | throw pillow | Textile | 96% | cream/beige | textured fabric | contemporary | good | small |
| 7 | 1 | throw pillow | Textile | 95% | cream/beige | textured fabric | contemporary | good | small |
| 8 | 1 | throw blanket | Textile | 88% | cream/beige | knitted fabric | contemporary | good | small |
| 9 | 1 | coffee table | Furniture | 97% | dark brown | wood | traditional/transitional | good | mediu |
| 10 | 1 | decorative vase | Decor | 91% | gold | ceramic/metal | modern | good | small |
| 11 | 1 | decorative bowl | Decor | 89% | dark brown | wood | modern | good | small |
| 12 | 1 | book | Decor | 90% | white | paper | none | good | small |
| 13 | 1 | vase with flowers | Decor | 95% | pink/green | glass/ceramic | floral | good | mediu |
| 14 | 1 | circular dining table | Furniture | 97% | white/light wood | wood | modern | good | small |
| 15 | 1 | dining chair | Furniture | 96% | gold/natural wood | metal/wood | modern | good | mediu |
| 16 | 1 | dining chair | Furniture | 95% | gold/natural wood | metal/wood | modern | good | mediu |
| 17 | 1 | plant in pot | Plant | 90% | green | ceramic/plant | natural | good | small |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 18 | 1 | plant in pot | Plant | 88% | green | ceramic/plant | natural | good | small |
| 19 | 1 | chandelier | Lighting | 95% | gold/clear | metal/glass | modern/art deco | good | mediu |
| 20 | 1 | floor lamp | Lighting | 94% | orange/gold | metal | mid-century modern | good | mediu |
| 21 | 1 | area rug | Textile | 97% | red/grey | wool | distressed oriental | good | large |
| 22 | 1 | art print | Art | 96% | red/orange | canvas/paper | abstract figurative | good | mediu |
| 23 | 1 | artwork | Art | 95% | monochrome | canvas | abstract | good | large |
| 24 | 1 | sconce | Lighting | 93% | gold | metal | modern | good | small |
| 25 | 1 | sconce | Lighting | 92% | gold | metal | modern | good | small |
| 26 | 1 | sideboard/console table | Furniture | 96% | gold/dark wood | metal/wood | traditional/glam | good | large |
| 27 | 1 | dining chair | Furniture | 97% | red/burgundy | velvet | mid-century modern | good | mediu |
| 28 | 1 | dining chair | Furniture | 96% | red/burgundy | velvet | mid-century modern | good | mediu |
| 29 | 1 | rug/sheepskin | Textile | 90% | white | faux fur | bohemian | good | small |
| 30 | 1 | plant | Plant | 95% | green | plant | natural | good | large |
| 31 | 1 | artwork | Art | 90% | dark | canvas/paper | minimalist | good | small |
| 32 | 1 | artwork | Art | 88% | dark | canvas/paper | minimalist | good | small |
| 33 | 1 | table lamp | Lighting | 88% | gold | metal | modern | good | small |
| 34 | 1 | dining table | Furniture | 90% | gold | metal/glass | glam | good | mediu |
| 35 | 1 | chair | Furniture | 92% | black | wood | traditional | good | mediu |
| 36 | 2 | decorative vase | Decor | 85% | gold | ceramic | modern | good | small |
| 37 | 2 | small sculpture | Decor | 80% | dark | unknown | abstract | good | very small |
| 38 | 2 | decorative tray | Storage | 78% | gold | metal | modern | good | small |
| 39 | 2 | table lamp | Lighting | 88% | orange | glass | mid-century modern | good | mediu |
| 40 | 2 | side table | Furniture | 82% | black | wood | minimalist | good | small |
| 41 | 2 | books | Decor | 80% | various | paper | mixed | good | mediu |
| 42 | 2 | table coaster | Accessory | 75% | brown | leather | modern | good | small |
| 43 | 2 | decorative vase | Decor | 85% | pink | ceramic | modern | good | small |
| 44 | 2 | flowers | Plant | 90% | pink | natural | fresh | good | small |
| 45 | 2 | ottoman/pouf | Furniture | 70% | dark red | fabric | modern | good | small |
| 46 | 2 | candle holder | Decor | 80% | black | metal | modern | good | very small |
| 47 | 2 | wine glass | Tableware | 70% | clear | glass | stemmed | good | small |
| 48 | 2 | wine glass | Tableware | 68% | clear | glass | stemmed | good | small |
| 49 | 2 | coaster | Accessory | 60% | grey | felt | round | good | very small |
| 50 | 2 | framed photo | Art | 75% | gold | metal | classic | good | small |
| 51 | 2 | framed photo | Art | 70% | gold | metal | classic | good | small |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 52 | 2 | wall sconce | Lighting | 88% | gold | metal | modern | good | small |
| 53 | 2 | wall sconce | Lighting | 85% | gold | metal | modern | good | small |
| 54 | 2 | wall sconce | Lighting | 80% | gold | metal | modern | good | small |
| 55 | 2 | picture frame | Art | 90% | black | wood | minimalist | good | large |
| 56 | 2 | picture frame | Art | 88% | black | wood | minimalist | good | mediu |

SECOND PASS

# 21 Newly Found Items

#36 decorative vase (Decor, 85%) — Small item partially obscured on a table.
#37 small sculpture (Decor, 80%) — Very small item on a coffee table, easily overlooked.
#38 decorative tray (Storage, 78%) — Partially obscured by other items on the coffee table.
#39 table lamp (Lighting, 88%) — Partially hidden behind the sectional sofa.
#40 side table (Furniture, 82%) — Partially obscured by the chair in the foreground.
#41 books (Decor, 80%) — Small items on the coffee table, clustered together.
#42 table coaster (Accessory, 75%) — Very small item on the coffee table.
#43 decorative vase (Decor, 85%) — Small item on the coffee table.
#44 flowers (Plant, 90%) — Separate from their vase, could be overlooked if only looking for 'vase with flowers'.
#45 ottoman/pouf (Furniture, 70%) — Partially cut off at the edge of the frame.
#46 candle holder (Decor, 80%) — Small item on the coffee table, easily obscured.
#47 wine glass (Tableware, 70%) — Small item on the dining table, partially obscured.
#48 wine glass (Tableware, 68%) — Small item on the dining table, partially obscured.
#49 coaster (Accessory, 60%) — Very small item on the dining table, easily overlooked.
#50 framed photo (Art, 75%) — Small item on the wall, partially obscured by the larger artwork.
#51 framed photo (Art, 70%) — Small item on the wall, partially obscured by the larger artwork.
#52 wall sconce (Lighting, 88%) — Part of a decorative cluster, could be mistaken as part of the artwork.
#53 wall sconce (Lighting, 85%) — Part of a decorative cluster, could be mistaken as part of the artwork.
#54 wall sconce (Lighting, 80%) — Partially obscured by the wall and only a section is visible.
#55 picture frame (Art, 90%) — Often overlooked if the 'artwork' itself was detected.
#56 picture frame (Art, 88%) — Often overlooked if the 'artwork' itself was detected.

ARCHITECTURE

# System Components

**Base Model:** Qwen2.5-VL-3B-Instruct — chosen for its native bounding-box grounding, JSON-structured output support, and 3B parameter size (fits on a single consumer GPU with QLoRA).

**Fine-tuning Method:** QLoRA (4-bit quantization + Low-Rank Adaptation)

Only 0.5-2% of parameters are trainable (LoRA adapters)
12GB VRAM minimum (RTX 3060/4070), 24GB recommended (RTX 4090)
Training time: ~30 min per epoch on 100 images (single GPU)

**Training Data Format:** Chat-based conversations matching Qwen2.5-VL's native format — system prompt sets the inventory task, user provides the image, assistant provides the JSON ground truth.

WORKFLOW

# Continuous Learning Loop

**Phase 1 — Cold Start (Bootstrap)**

Collect room photos (20-50 images minimum)
Auto-label using a large VLM API (GPT-4o, Gemini)
Human reviews and corrects labels
First training run produces v1 adapter

**Phase 2 — Active Learning Cycle**

New images arrive (user uploads, batch import)
Current fine-tuned model auto-labels them
Human reviews corrections (focus on errors)
When 20+ reviewed images accumulate, retrain

Evaluate new model on held-out validation set
If F1 improves, promote to production; if not, keep previous

**Phase 3 — Specialization**

Model becomes increasingly accurate on YOUR specific home style, furniture brands, decor patterns
Fewer corrections needed over time
Can specialize further: room-type-specific adapters, seasonal inventory tracking

DATASET

# Data Management

The `DatasetManager` handles:

- **Deduplication** — SHA-256 hashing prevents duplicate images
- **Versioning** — manifest history tracks every change
- **Train/Val splits** — reviewed (corrected) images always go to training
- **COCO-compatible export** — for evaluation with standard tools

**Dataset structure:**

```
dataset/
  images/           # deduplicated room photos
  labels/           # JSON ground truth per image
  manifest.json     # master index
  splits.json       # train/val assignments
  history/          # manifest versions
```

EVALUATION

# Metrics & Quality

The evaluation system measures:

- **mAP @ IoU 0.5** — standard object detection metric (bounding box accuracy)
- **Per-category precision/recall** — which object types the model handles well
- **Tag accuracy** — color, material, style, condition match rates
- **JSON parse rate** — does the model produce valid structured output?

Training history tracks all versions, so you can see improvement over time and roll back if a training run regresses.

QUICK START

# Getting Started

BASH

```bash
# 1. Install dependencies
pip install torch torchvision transformers peft bitsandbytes
pip install accelerate datasets pillow qwen-vl-utils trl flash-attn

# 2. Bootstrap dataset from your room photos
python home_inventory_training.py bootstrap \
    --images ./room_photos/ \
    --dataset ./dataset/ \
    --api-key sk-... \
    --provider openrouter

# 3. Review labels, correct any errors, then train
python home_inventory_training.py train \
    --dataset ./dataset/ \
    --output ./checkpoints/ \
    --epochs 5 \
    --lr 1e-5

# 4. As new images come in, ingest and retrain
python home_inventory_training.py ingest \
    --images new_room1.jpg new_room2.jpg \
    --dataset ./dataset/ \
    --checkpoints ./checkpoints/
```

PYTHON API

## Programmatic Usage

PYTHON

```python
from home_inventory_training import (
    ContinuousTrainingOrchestrator,
    FineTunedInventoryModel,
    TrainingConfig,
)

# Set up the orchestrator
orch = ContinuousTrainingOrchestrator(
    dataset_dir="./dataset",
    checkpoints_dir="./checkpoints",
    api_key="sk-...",
    retrain_threshold=20,  # retrain after 20 new reviewed images
    config=TrainingConfig(
        num_epochs=5,
        learning_rate=1e-5,
        lora_r=16,
    ),
)

# Ingest new room photos
orch.ingest_images(["kitchen.jpg", "bedroom.jpg", "office.jpg"])

# After human review, check if retraining is due
if orch.should_retrain():
    metrics = orch.retrain()
    print(f"New model F1: {metrics['f1']}")

# Use the fine-tuned model for inference
model = FineTunedInventoryModel("./checkpoints/v1/final/")
inventory = model.detect("new_room.jpg")
print(json.dumps(inventory, indent=2))
```